

4.5 The Cochran theorem

An analysis of variance starts from an identity

$$\mathbf{Y}^\top \mathbf{Y} = \mathbf{Y}^\top \mathbf{A}_1 \mathbf{Y} + \cdots + \mathbf{Y}^\top \mathbf{A}_k \mathbf{Y}, \quad \mathbf{A}_1 + \cdots + \mathbf{A}_k = \mathbf{I},$$

that splits the total sum of squares into pieces attributed to sources of variation. Theorem 4.4.4 says the pieces are independent chi-squared variables when the $\mathbf{A}_i$ are mutually orthogonal projections. Checking that directly means checking k idempotence conditions and $\binom{k}{2}$ orthogonality conditions. Cochran's theorem shows that one count of ranks does all of it.

4.5.1 The algebra

The matrix algebra behind this was done in [Chapter 1](#). For symmetric matrices, any two of the three properties “each $\mathbf{A}_i$ is idempotent”, “ $\mathbf{A}_i \mathbf{A}_j = \mathbf{0}$ for $i \neq j$ ” and “the sum is idempotent” imply the third, and symmetric matrices that add to $\mathbf{I}$ with ranks adding to n are idempotent and mutually orthogonal (Theorem 1.9.3). The next theorem adds the rank count for a sum that is idempotent but not necessarily $\mathbf{I}$, which is what nested models need.

Theorem 4.5.1 — Cochran, algebraic form

Let $\mathbf{A}_1, \dots, \mathbf{A}_k$ be symmetric $n \times n$ matrices and $\mathbf{A} = \sum_i \mathbf{A}_i$. Consider

- (a) each $\mathbf{A}_i$ is idempotent;
- (b) $\mathbf{A}_i \mathbf{A}_j = \mathbf{0}$ for all $i \neq j$;
- (c) $\mathbf{A}$ is idempotent;
- (d) $\text{rank}(\mathbf{A}) = \sum_i \text{rank}(\mathbf{A}_i)$.

Then any two of (a), (b), (c) imply all four, and (c) together with (d) implies all four.

Proof. Any two of (a), (b), (c). By Theorem 1.9.3, any two of them imply the third. When all three hold, rank equals trace for symmetric idempotent matrices (Theorem 1.9.2), so $\text{rank}(\mathbf{A}) = \text{tr}(\mathbf{A}) = \sum_i \text{tr}(\mathbf{A}_i) = \sum_i \text{rank}(\mathbf{A}_i)$, which is (d).

(c) and (d) imply (a) and (b). Let $r = \text{rank}(\mathbf{A})$ and $r_i = \text{rank}(\mathbf{A}_i)$. Every $\mathbf{A}\mathbf{v} = \sum_i \mathbf{A}_i \mathbf{v}$ lies in $\mathcal{C}(\mathbf{A}_1) + \cdots + \mathcal{C}(\mathbf{A}_k)$, whose dimension is at most $\sum_i r_i = r = \dim \mathcal{C}(\mathbf{A})$. So the two spaces coincide, and in particular $\mathcal{C}(\mathbf{A}_i) \subseteq \mathcal{C}(\mathbf{A})$. By Theorem 1.9.2(a), $\mathbf{A} = \mathbf{Q}\mathbf{Q}^\top$ with $\mathbf{Q}$ of size $n \times r$ and $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_r$. Since $\mathbf{A}$ is symmetric and idempotent it fixes every vector of its column space, so $\mathbf{A}\mathbf{A}_i = \mathbf{A}_i$, and transposing, $\mathbf{A}_i \mathbf{A} = \mathbf{A}_i$. Hence $\mathbf{A}_i = \mathbf{A}\mathbf{A}_i \mathbf{A} = \mathbf{Q}\mathbf{B}_i \mathbf{Q}^\top$ with $\mathbf{B}_i = \mathbf{Q}^\top \mathbf{A}_i \mathbf{Q}$. The $r \times r$ matrices $\mathbf{B}_i$ are symmetric, satisfy $\sum_i \mathbf{B}_i = \mathbf{Q}^\top \mathbf{A} \mathbf{Q} = \mathbf{I}_r$, and have $\text{rank}(\mathbf{B}_i) = r_i$ because $\mathbf{Q}$ has orthonormal columns. By the last statement of Theorem 1.9.3, $\mathbf{B}_i^2 = \mathbf{B}_i$ and $\mathbf{B}_i \mathbf{B}_j = \mathbf{0}$ for $i \neq j$, and multiplying by $\mathbf{Q}$ on the left and $\mathbf{Q}^\top$ on the right transfers both identities to the $\mathbf{A}_i$. ■

4.5.2 The distributional statement

Theorem 4.5.2 — Cochran

Let $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$, and let $\mathbf{A}_1, \dots, \mathbf{A}_k$ be symmetric with ranks $r_1, \dots, r_k$, whose sum $\mathbf{A} = \sum_i \mathbf{A}_i$ is idempotent of rank r (for example $\mathbf{A} = \mathbf{I}$, $r = n$). The following are equivalent:

- (i) the forms $\mathbf{Y}^\top \mathbf{A}_i \mathbf{Y} / \sigma^2$ are mutually independent, and $\mathbf{Y}^\top \mathbf{A}_i \mathbf{Y} / \sigma^2 \sim \chi^2(r_i, \boldsymbol{\mu}^\top \mathbf{A}_i \boldsymbol{\mu} / \sigma^2)$ for each i ;
- (ii) $r_1 + \dots + r_k = r$;
- (iii) each $\mathbf{A}_i$ is idempotent;
- (iv) $\mathbf{A}_i \mathbf{A}_j = \mathbf{0}$ for all $i \neq j$.

Proof. Condition (c) of Theorem 4.5.1 holds by assumption, so that theorem shows that (ii), (iii) and (iv) are equivalent, and each implies the other two. If (iii) and (iv) hold, Theorem 4.4.4 gives (i). If (i) holds, apply Theorem 4.3.1 to $\sigma^{-1} \mathbf{Y} \sim N_n(\sigma^{-1} \boldsymbol{\mu}, \mathbf{I})$. Each $\mathbf{A}_i$ with $r_i \geq 1$ is idempotent, and an $\mathbf{A}_i$ of rank zero is $\mathbf{0}$. This is (iii). ■

In words: *if the degrees of freedom add up, the sums of squares are independent chi-squared variables.* Independence in (i) is not needed to reach (iii). The marginal chi-squared laws alone force the whole structure.

Example. (The one-way layout)

Let observations fall into g groups of sizes $n_1, \dots, n_g$, with $Y_{\ell j}$ independent $N(\mu_\ell, \sigma^2)$. Let $\mathbf{M}$ replace each observation by its group mean, and $\mathbf{P}_1 = n^{-1} \mathbf{1} \mathbf{1}^\top$ by the grand mean. Then

$$\mathbf{I} = \mathbf{P}_1 + (\mathbf{M} - \mathbf{P}_1) + (\mathbf{I} - \mathbf{M})$$

splits $\|\mathbf{Y}\|^2$ into $n\bar{Y}^2$, the between-group sum of squares $\sum_\ell n_\ell (\bar{Y}_\ell - \bar{Y})^2$ and the within-group sum of squares $\sum_{\ell,j} (Y_{\ell j} - \bar{Y}_\ell)^2$. The three matrices are symmetric with ranks 1, $g - 1$ and $n - g$, which add to n . Cochran's theorem needs nothing more. The three sums of squares, divided by σ^2 , are independent noncentral chi-squared variables. Their noncentralities are $n\bar{\mu}^2/\sigma^2$, $\sum_\ell n_\ell (\mu_\ell - \bar{\mu})^2/\sigma^2$ with $\bar{\mu} = \sum_\ell n_\ell \mu_\ell / n$, and 0. The last is zero because $\mathbf{M}\boldsymbol{\mu} = \boldsymbol{\mu}$. The ratio of the between and within mean squares is $F(g - 1, n - g, \gamma)$ by Theorem 4.4.6, and it is central iff all group means are equal.

The script uses $g = 3$ groups of sizes 3, 5, 4, means 2, 3, 1.5, and $\sigma = 1$, and draws 200,000 data sets. Table 4.5.1 compares the theoretical means $r_i + \gamma_i$ and variances $2r_i + 4\gamma_i$ with the simulated ones. The largest absolute correlation between the three sums of squares is 0.0013. As a check that goes beyond correlation, the proportion of data sets in which both the between and the within sums of squares exceed their medians is 0.2497, against 0.25 for independent variables.

```
import numpy as np
from scipy import stats
rng = np.random.default_rng(5150)

sizes, means, sigma = [3, 5, 4], [2.0, 3.0, 1.5], 1.0
groups = np.repeat(np.arange(3), sizes)
```

```

n = len(groups)
Z = (groups[:, None] == np.arange(3)).astype(float)      # group indicators
M = Z @ np.diag(1 / np.array(sizes)) @ Z.T               # group-mean projection
J = np.ones((n, n)) / n
pieces = [J, M - J, np.eye(n) - M]
ranks = [np.linalg.matrix_rank(P) for P in pieces]
print("ranks", ranks, "sum", sum(ranks), "n", n)

mu = np.array(means)[groups]
gammas = [mu @ P @ mu / sigma**2 for P in pieces]
Y = mu + sigma * rng.standard_normal((200_000, n))
Q = np.stack([np.einsum("ij,jk,ik->i", Y, P, Y) for P in pieces], axis=1) / sigma**2
print(np.corrcoef(Q, rowvar=False).round(3))

```

[Run this code online](#)

Table 4.5.1. Cochran decomposition for a one-way layout with $n = 12$: theory against 200,000 simulated data sets.

Sum of squares	r_i	γ_i	mean	simulated	variance	simulated
grand mean	1	60.750	61.750	61.763	245.00	244.95
between groups	2	5.250	7.250	7.230	25.00	24.85
within groups	9	0.000	9.000	8.994	18.00	17.96

Remark (What the theorem does not cover). Two assumptions carry the weight. The first is normality. Without it, Theorem 4.2.3(d) fails, and with independent non-normal errors the between and within sums of squares are in general correlated, with a correlation driven by the fourth cumulant (Exercise (B2)). The second is the spherical covariance $\sigma^2 \mathbf{I}$. With $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{V}$, $\mathbf{V}$ positive definite, apply the theorem to $\mathbf{V}^{-1/2} \mathbf{Y}$, whose forms have matrices $\mathbf{V}^{1/2} \mathbf{A}_i \mathbf{V}^{1/2}$. Their ranks are those of the $\mathbf{A}_i$, so the rank count is unchanged. What changes is the condition on the sum: $\mathbf{V}^{1/2} \mathbf{A} \mathbf{V}^{1/2}$ is idempotent iff $\mathbf{A} \mathbf{V} \mathbf{A} = \mathbf{A}$ (for example $\sum_i \mathbf{A}_i = \mathbf{V}^{-1}$), and conditions (iii) and (iv) become $\mathbf{A}_i \mathbf{V} \mathbf{A}_i = \mathbf{A}_i$ and $\mathbf{A}_i \mathbf{V} \mathbf{A}_j = \mathbf{0}$. These are the generalized least squares decompositions of [Chapter 31](#) (Theorem 31.1.1).

Most analysis of variance tables in this book come from a chain of nested model spaces. The successive differences of their projections are mutually orthogonal projections whose ranks add up, so Cochran's theorem applies without further work ([Exercise \(B2\)](#), Theorem 9.1.1).

4.5.3 Exercises

4.5.3.1 A. Check your understanding

Exercise A1.

Let $\mathbf{A}_1 = \text{diag}(2, 0)$ and $\mathbf{A}_2 = \text{diag}(-1, 1)$, so that $\mathbf{A}_1 + \mathbf{A}_2 = \mathbf{I}_2$. Which hypotheses of [Theorem 4.5.2](#) fail, and what are the laws of the two forms when $\mathbf{Y} \sim N_2(\mathbf{0}, \mathbf{I})$?

Solution. The sum $\mathbf{I}_2$ is idempotent, but every one of the conditions (ii)–(iv) fails. The ranks are 1 and 2, which add to $3 \neq 2$. Neither matrix is idempotent, since $\mathbf{A}_1^2 = \text{diag}(4, 0)$ and $\mathbf{A}_2^2 = \mathbf{I}_2$. And $\mathbf{A}_1\mathbf{A}_2 = \text{diag}(-2, 0) \neq \mathbf{0}$. The forms are $2Y_1^2$, which is twice a $\chi^2(1)$ variable, and $Y_2^2 - Y_1^2$, the difference of two independent $\chi^2(1)$ variables, which takes negative values. Neither is chi-squared. They are dependent, because

$$\text{Cov}(2Y_1^2, Y_2^2 - Y_1^2) = 2 \text{Cov}(Y_1^2, Y_2^2) - 2 \text{Var}(Y_1^2) = 0 - 2 \cdot 2 = -4 \neq 0.$$

4.5.3.2 B. Practice

Exercise B1.

A two-way layout has one observation Y_{ij} for each of a rows and b columns. Write $\|\mathbf{Y}\|^2$ as the sum of the grand-mean, row, column and residual sums of squares, identify the four matrices, and use [Theorem 4.5.2](#) to show that under $Y_{ij} \sim N(\mu + \alpha_i + \beta_j, \sigma^2)$ independently the four sums of squares are independent, with degrees of freedom 1, $a - 1$, $b - 1$ and $(a - 1)(b - 1)$.

Exercise B2.

Let $\mathbf{P}_1, \dots, \mathbf{P}_m$ be symmetric idempotent with $\mathcal{C}(\mathbf{P}_1) \subset \mathcal{C}(\mathbf{P}_2) \subset \dots \subset \mathcal{C}(\mathbf{P}_m)$, and put $\mathbf{P}_0 = \mathbf{0}$, $\mathbf{P}_{m+1} = \mathbf{I}$. Show that the matrices $\mathbf{P}_i - \mathbf{P}_{i-1}$, $i = 1, \dots, m + 1$, satisfy the conditions of [Theorem 4.5.2](#), so that the sequential sums of squares $\|(\mathbf{P}_i - \mathbf{P}_{i-1})\mathbf{Y}\|^2$ are independent noncentral chi-squared variables when $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$.

4.5.3.3 C. Going deeper

Exercise C1.

Let $\mathbf{Y} \sim N_n(\mathbf{0}, \mathbf{I})$, and $Q = Q_1 + Q_2$ with $Q = \mathbf{Y}^\top \mathbf{A} \mathbf{Y} \sim \chi^2(r)$, $Q_1 = \mathbf{Y}^\top \mathbf{A}_1 \mathbf{Y} \sim \chi^2(r_1)$ and $Q_2 = \mathbf{Y}^\top \mathbf{A}_2 \mathbf{Y}$ for a nonnegative definite $\mathbf{A}_2$. Show that Q_1 and Q_2 are independent and $Q_2 \sim \chi^2(r - r_1)$.

Solution. By Theorem 4.3.1, $\mathbf{A}$ and $\mathbf{A}_1$ are idempotent, with ranks r and r_1 . If $\mathbf{A}\mathbf{v} = \mathbf{0}$, then $0 = \mathbf{v}^\top \mathbf{A}_1 \mathbf{v} + \mathbf{v}^\top \mathbf{A}_2 \mathbf{v}$ with both terms nonnegative, so $\|\mathbf{A}_1 \mathbf{v}\|^2 = \mathbf{v}^\top \mathbf{A}_1 \mathbf{v} = 0$. Thus $\mathcal{N}(\mathbf{A}) \subseteq \mathcal{N}(\mathbf{A}_1)$, hence $\mathcal{C}(\mathbf{A}_1) \subseteq \mathcal{C}(\mathbf{A})$ and $\mathbf{A}\mathbf{A}_1 = \mathbf{A}_1 = \mathbf{A}_1\mathbf{A}$. Then $\mathbf{A}_2^2 = (\mathbf{A} - \mathbf{A}_1)^2 = \mathbf{A} - 2\mathbf{A}_1 + \mathbf{A}_1 = \mathbf{A}_2$, so $\mathbf{A}_2$ is idempotent, and $\mathbf{A}_1\mathbf{A}_2 = \mathbf{A}_1\mathbf{A} - \mathbf{A}_1 = \mathbf{0}$. By Theorem 4.4.4, Q_1 and Q_2 are independent and $Q_2 \sim \chi^2(\text{tr } \mathbf{A}_2) = \chi^2(r - r_1)$.